The first criterion of congruence of triangles

Two sides and the angle between them fix a triangle completely.

LESSON 47–48 | 2 × 40 MIN | MATEMATIKA 7, §20 | S8 5

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

What congruence means	15 min
The criterion	25 min
Writing a proof	25 min
Consequences	20 min
Homework	5 min

LEARNING OBJECTIVES

- Define congruent triangles and use the symbol ≡.
- State and apply the SAS criterion.
- Write a proof of congruence in statement-and-reason form.
- Deduce equal sides and angles from a congruence.

TEXTBOOK

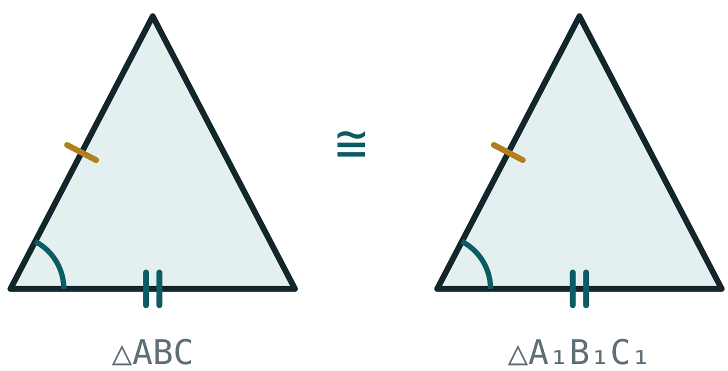
National	pp. 123–129
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01 Explanation

What congruence means

Two triangles are **congruent** if one can be laid exactly on the other. Written $\triangle ABC \equiv \triangle A_1B_1C_1$, with the vertices in matching order.

Congruent triangles have **six** pairs of equal elements: three sides and three angles.



Congruent triangles — same shape and same size, in any position.

THE ORDER OF THE LETTERS IS PART OF THE STATEMENT

$\triangle ABC \equiv \triangle A_1B_1C_1$ says A matches A_1, B matches B_1 and C matches C_1. Writing them in the wrong order claims something false.

The criterion

> If two sides of one triangle and the angle between them are equal to two sides and the angle between them of another, the triangles are congruent

$$AB = A_1B_1,\ AC = A_1C_1,\ \angle A = \angle A_1 \Longrightarrow \triangle ABC \equiv \triangle A_1B_1C_1$$

Three of the six pairs are enough — provided they are the **right** three.

GIVEN	CONGRUENT?
two sides and the included angle	yes — the first criterion
two sides and a non-included angle	not necessarily
three angles	no — only similar

WHY THE ANGLE MUST BE INCLUDED

Two sides fixed at a hinge open and close: the third side changes with the angle. Fixing that angle fixes the triangle. With a non-included angle the hinge is not fixed and two shapes remain possible.

Writing a proof

Problem. $AB = AD$ and $\angle BAC = \angle DAC$. Prove $\triangle ABC \equiv \triangle ADC$.

STATEMENT	REASON
$AB = AD$	given
$\angle BAC = \angle DAC$	given
$AC = AC$	common side
$\triangle ABC \equiv \triangle ADC$	first criterion (SAS)

THE COMMON SIDE IS THE MARK MOST OFTEN MISSED

Almost every proof of this kind needs it, and it is always available. Looking for a shared side or a shared angle is the first thing to do in any congruence question.

Consequences

Once a congruence is proved, **all six** pairs of elements are equal. That is what makes the criteria useful: three facts are given, and three more follow free.

PROVED	THEN ALSO
$\triangle ABC \equiv \triangle ADC$	$BC = DC$
same	$\angle B = \angle D$
same	$\angle BCA = \angle DCA$

The abbreviation used in proofs is *CPCT* — corresponding parts of congruent triangles.

MATCH THE ELEMENTS BY POSITION, NOT BY APPEARANCE

In $\triangle ABC \equiv \triangle ADC$, BC corresponds to DC because B matches D and C matches C. Reading the correspondence off the letters is always safe; reading it off the drawing is not.

02 Worked examples

EXAMPLE 1 **$AB = AD$ and $\angle BAC = \angle DAC$. Prove $\triangle ABC \equiv \triangle ADC$.**

1. $AB = AD$ — given.
2. $\angle BAC = \angle DAC$ — given.
   The included angle.
3. AC is common to both.
   The extra fact.
4. By the first criterion, $\triangle ABC \equiv \triangle ADC$.

ANSWER

Proved by SAS

EXAMPLE 2 **In the same figure, what else follows?**

1. All six pairs of elements are equal.
   CPCT.
2. $BC = DC$
3. $\angle B = \angle D$
4. $\angle BCA = \angle DCA$

ANSWER

$BC = DC$, $\angle B = \angle D$, $\angle BCA = \angle DCA$

EXAMPLE 3 **Two triangles have sides 6 and 8 with an angle of $40°$. Must they be congruent?**

1. If the $40°$ lies between the 6 and the 8: yes.
   SAS.
2. If it lies elsewhere: not necessarily.
3. The position of the angle decides.
4. Only when it is the included angle.

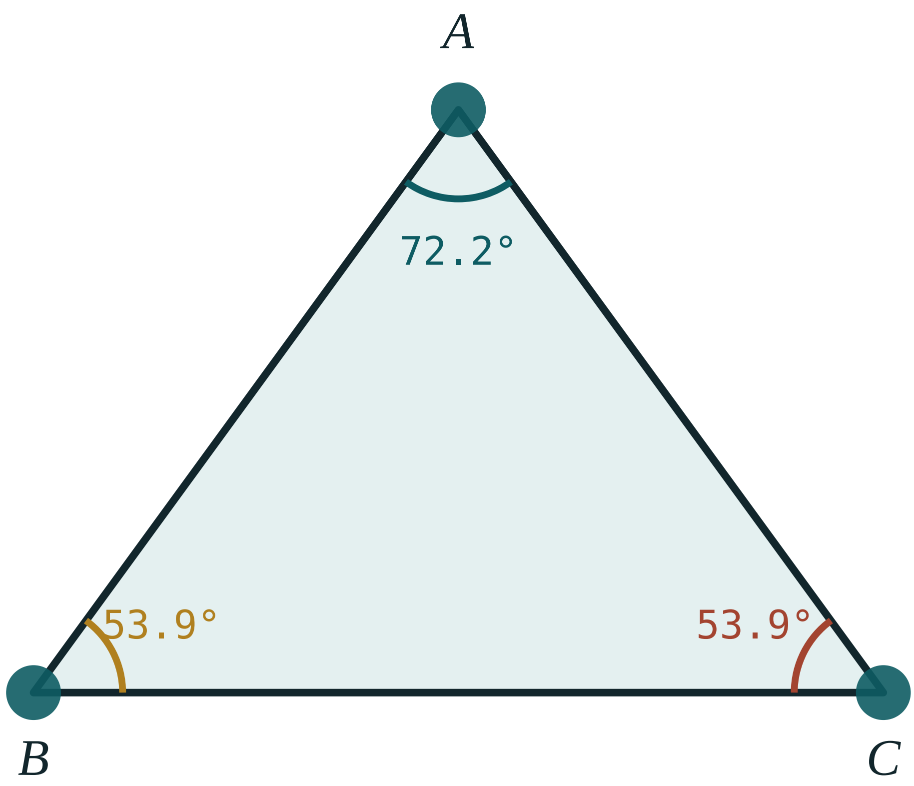
ANSWER

Only if the angle is included

03 Interactive model

Hinge two rulers at a fixed angle and mark the third side; changing the angle changes it, and the criterion becomes a physical fact.

Two sides and the angle between



∠A **72.2°** | ∠B **53.9°** | ∠C **53.9°** | sum **180°** | longest side **BC** | largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Congruent	Teng	Равный
Criterion	Alomat	Признак
Corresponding elements	Mos elementlar	Соответственные элементы
Included angle	Orasidagi burchak	Угол между сторонами
Superposition	Ustma-ust qo‘yish	Наложение
To follow	Kelib chiqmoq	Следовать
Common side	Umumiy tomon	Общая сторона
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Congruent triangles have how many pairs of equal elements?

- *3*
- *4*
- *6*
- *2*

The first criterion needs:

- three sides
- two angles
- two sides and the included angle
- three angles

Three equal angles give:

- congruence
- similarity only
- nothing
- equal areas

A shared side in a figure is:

- irrelevant
- a usable equality
- a mistake
- an angle

CPCT means:

- a criterion
- corresponding parts of congruent triangles
- a construction
- a theorem name

A non-included equal angle gives:

- congruence
- not necessarily congruence
- similarity only
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Congruent triangles have how many equal pairs?

   ANSWER 6

2. The symbol for congruence

   ANSWER ≡

3. The first criterion in letters

   ANSWER SAS

4. A common side gives

   ANSWER An equality for free

5. Three equal angles give

   ANSWER Similarity only

6. In $\triangle ABC \equiv \triangle DEF$, AB matches

   ANSWER DE

7. In the same, $\angle C$ matches

   ANSWER $\angle F$

Medium · the standard question

7 PROBLEMS

1. $AB = AD$, $\angle BAC = \angle DAC$: the third fact

   ANSWER AC common

2. What follows about BC?

   ANSWER $BC = DC$

3. What follows about $\angle B$?

   ANSWER $\angle B = \angle D$

4. Sides 6, 8 with $40°$ included: congruent?

   ANSWER Yes

5. Sides 6, 8 with $40°$ not included: congruent?

   ANSWER Not necessarily

6. Two triangles with 5, 7 and $60°$ included

   ANSWER Congruent

7. Are congruent triangles similar?

   ANSWER Yes, with $k = 1$

Hard · stretch and reason

7 PROBLEMS

1. In a parallelogram, prove the two triangles cut by a diagonal are congruent

   ANSWER SAS with the common diagonal

2. Prove that the diagonals of a parallelogram bisect each other

   ANSWER Congruent triangles at the centre

3. An isosceles triangle: prove the bisector from the apex is a median

   ANSWER SAS

4. Two triangles with $AB = DE, AC = DF, \angle A = \angle D$: name the criterion

   ANSWER The first

5. Why is AAA not enough?

   ANSWER Any enlargement has the same angles

6. In $\triangle ABC \equiv \triangle DEF$ with $AB = 5, \angle B = 40°$: find DE and $\angle E$

   ANSWER 5 and $40°$

7. Prove that a kite has two pairs of congruent triangles

   ANSWER SAS on each diagonal

07 Homework

Homework — 5 tasks

Look for a common side or a common angle in every figure before starting.

1. In a figure with $AB = AC$ and $\angle BAD = \angle CAD$, prove $\triangle ABD \equiv \triangle ACD$.
2. State what else follows from that congruence.
3. Explain why the equal angle must lie between the two equal sides.
4. Two triangles have sides 7 and 9 with an included angle of $55°$. Are they congruent?
5. Prove that a diagonal of a parallelogram divides it into two congruent triangles.